Ahmed Younis at ArGuard Shared Tasks: Calibration Is Most of What You Tune
Decode-Time Structure and Threshold Transport for Arabic Harmful-Content Detection

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Abstract

We describe our systems for all four ArGuard 2026 subtasks: binary and fine-grained Arabic hateful-meme detection (A1/A2) and binary and granular harmful-prompt detection (B1/B2). Our systems are deliberately conventional (a LoRA-tuned Qwen3-VL-8B for memes, a nine-encoder ensemble for prompts) because our central finding is that most of the achievable headroom lay not in the model but in decode-time calibration: per-class thresholds refit under structural constraints, rank-based decoding against known or published class priors, and threshold transport under distribution shift. On subtask A2, our submissions rose from 0.248 to 0.419 macro-F1; roughly 0.15 of that gain came from decode-time changes against under 0.03 from model-side interventions, and a fixed-model ablation on the ensemble's char-ngram member attributes +0.070 to the decode stack alone. On B1, recalibrating a single threshold recovered +0.16 macro-F1 after we detected that the blind test was rebalanced to ≈49% unsafe against an 80% unsafe training prior. Our final systems ranked **1st on subtask A2** (macro-F1 0.4194; 0.3913 without per-class scoring feedback), 3rd on A1 (0.7889), 3rd on B2 (0.7695), and 7th on B1 (0.7552). We additionally report a benchmark audit conducted during participation: a sequential-id label leak in the Task B development release (macro-F1 1.000 from ids alone), identified before the test release, left unexploited, and absent from the re-indexed test files; latent taxonomy structure that contradicts the task description; and a quantified gap between development and test distributions.

1 Introduction

The ArGuard shared task at ArabicNLP 2026 (Alam et al., 2026) targets two Arabic safety problems: multimodal hateful-meme detection over the AHA-Memes corpus (Kmainasi et al., 2026) (subtasks A1 binary, A2 fine-grained multi-label) and harmful-prompt detection for LLMs (B1 binary safe/unsafe, B2 seven-way harm domain). All subtasks rank by macro-F1.[1]

Shared-task papers usually foreground architecture. Our experience ran the other way: with competent but unremarkable models, the score was dominated by *decisions made after the forward pass*: which thresholds, fitted on which scores, against which prior. We therefore organize this paper around three calibration mechanisms and one benchmark audit:

- **Structure-aware threshold refitting** (A2): the fine-grained label space is deterministically partitioned by the binary label; exploiting this helps *only* if thresholds are refit on the constrained scores (§4.1).
- **Rank decoding against known priors** (A1/A2): the dataset card publishes the blind-test class balance; per-class scoring feedback reveals support. Decoding by rank against these counts replaces noisy threshold fits (§4.2).
- **Threshold transport under shift** (B1): the blind test was rebalanced relative to train; we show the shift is a threshold problem, not a representation problem (§4.3).
- **A benchmark audit** (§6): a complete label leak in the Task B development ids, identified pre-test-release, left unexploited, and absent from the re-indexed test files; latent taxonomy structure (§2); and development-set inflation from near-duplicate construction.

2 Data and Preprocessing

Task A (memes). We used only the official AHA-Memes release (Kmainasi et al., 2026): 3,500

[1]We competed as team *Ahmed Younis*, Codabench username `ahmedbakr`. Official final runs: A1 submission 881206, A2 881205, B1 881203, B2 881204. Code: `github.com/itsbakr/arguard-2026`.

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train / 500 dev / 500 dev-test / 500 blind test memes with OCR text. No external data: we measured the permitted alternatives and rejected them (Prop2Hate-Meme (Alam et al., 2024) is single-label LLM-silver with $\kappa \approx 0.4$ against humans and shares $\geq$10 images with ArGuard splits). Images were capped at 1024^2 pixels for both training and inference; the OCR `text` field, visibly noisy, was appended to the prompt as an auxiliary channel.

Task B (prompts). The official CSVs: 16,835 train + 2,405 dev rows (prompt, dialect, binary and granular labels). Two rows carry empty labels and were dropped. The dialect column, available at test time, was prepended as a literal token (e.g. `[Egyptian]`). We evaluated AraSafe (Mubarak et al., 2025) as permitted external data and measured zero prompt overlap with ArGuard but an inverted class balance and an incompatible taxonomy, so we did not use it for label supervision.

Three structural facts drove our design. (i) The A2 label space is partitioned by the A1 label: hateful memes carry only {Mocking, Incitement, Slurs, Dehumanization, Contempt, Inferiority, Exclusion, Other}; non-hateful only {Sarcasm, Humor, Other}. (ii) The B2 granular label populates *safe* rows as well (safe rows take only 4 of 7 categories, and `Bully`, `FraudandDeception` and `hate_speech` are 100% unsafe), so B2 is a seven-way harm-domain task over all rows, although the task description says "given an unsafe prompt". We verified against the platform: our all-rows offline scorer reproduced official scores to seven decimal places. (iii) The dataset card publishes the blind-test binary balance for Task A (148 Hateful / 352 Not Hateful; 29.6% vs. 37.8% in every public split).

3 System Architecture and Training

Task A: one vision-language model (VLM), two heads. We LoRA-tune (Hu et al., 2022) `QCRI/MemeLens-VLM` (Shahroor et al., 2026),[2] a Qwen3-VL-8B-Instruct (Qwen Team, 2025) checkpoint released by the organizers' group and already adapted to Arabic meme understanding, with r=32, α=64, dropout 0.05 on all LLM attention and MLP projections, vision tower frozen. MemeLens's own training mix includes Prop2Hate-Meme and ArMeme but not AHA-Memes, whose release postdates the checkpoint; exposure of the backbone to the small number of ArGuard images shared with Prop2Hate-Meme (§2), under a different taxonomy, therefore cannot be excluded. Rather than generative readout, a linear head over the last non-padding hidden state emits 1 binary + 10 sigmoid logits, trained jointly with class-weighted binary cross-entropy (per-class weight $\sqrt{\text{neg}/\text{pos}}$ capped at 8). lr 1e-4 cosine, effective batch 16, 3 epochs, bf16; $\sim$45 min per seed on one H100. Final submissions average probabilities over 8 seeds (5 trained on train, 3 on train+dev; thresholds for the train+dev seeds refit on out-of-fold scores).

Task B: encoder ensemble. Nine encoders (MARBERTv2 (Abdul-Mageed et al., 2021), AraBERTv02-twitter (Antoun et al., 2020), CAMeLBERT-mix (Inoue et al., 2021) $\times$ 3 seeds), each with a shared backbone and two heads (binary + seven-way) trained multi-task (lr 2e-5, 3 epochs, max_len 96); probabilities averaged. For B2 we blend at equal weight with a character 3–5-gram TF-IDF logistic model, which alone reaches dev macro-F1 0.86 and adds robustness to generation artifacts. B2 is decoded by argmax over the blended seven-way probabilities; its calibration leverage came upstream, from identifying that the metric is computed over all rows rather than only unsafe ones (§2), which changes both the training population and the class prior.

4 Decode-Time Calibration

4.1 Structure-aware refitting on A2

A2 macro-F1 averages ten *independent* binary columns (`zero_division=0`): a false positive on a rare class costs nothing elsewhere, each class is worth 0.1 of the metric, and an abstained class scores 0. A global 0.5 threshold therefore silently abandons the tail (Exclusion: 10 train positives). Table 1 shows our held-out ablation with the model fixed. Two results matter. First, per-class F1-optimal thresholds are worth +0.05. Second (the subtlety), gating scores by the A1 partition *alone hurts* (0.256$\rightarrow$0.197): the constraint shifts the score distribution and invalidates thresholds fitted on unconstrained scores. Refitting thresholds *on the gated scores* makes the same constraint profitable, and per-class floors at 1.5$\times$ the train prior recover rare-class recall. This decode stack took the identical model from 0.2559 to 0.3256 on held-out dev (+0.070). We frame the A1$\rightarrow$A2 constraint as hierarchical constrained decoding, following the label-hierarchy treatment of Dimitrov

[2] `huggingface.co/QCRI/MemeLens-VLM`

Decode rule (model fixed)	macro-F1
global 0.5 threshold	0.2559
global 0.5 + A1-gating	0.1972
per-class thresholds	0.3061
per-class, fit ungated, applied gated	0.2881
per-class, refit on gated scores	0.3123
+ rare-class floors (1.5× prior)	**0.3256**

Table 1: A2 decode ablation on held-out dev (char-ngram scores; identical ordering holds for the VLM). A structural constraint imposed without refitting *reduces* macro-F1.

et al. (2024).

4.2 Rank decoding against known counts

Where a class's true count is known or estimable, we replace thresholds with top-k rank decoding. For A1 the dataset card's published test balance (148/352) sets k: our submission at top-148 achieved hateful precision = recall = 0.703 exactly, confirming the published prior. For A2, per-class scoring feedback during the test phase reveals gold support; re-decoding against true support raised test macro-F1 from 0.3913 to 0.4167 in a single resubmission (Contempt, whose support the dataset card's union-of-matching-annotators test protocol inflated 2.6× relative to single-annotator dev, gained +0.247 F1). We discuss the propriety of this channel in §6.

4.3 Threshold transport under shift on B1

Our first B1 test submission, thresholded at the dev-optimal operating point, scored 0.5959, far below its dev value of 0.954. Backing the implied balance out of the platform's accuracy reading (0.6417 accuracy at 84.4% predicted unsafe) indicated the blind test is ≈49% unsafe versus 80% in train, assuming near-total unsafe recall at that operating point (which held on dev): a deliberate rebalance. Re-pinning the *same ensemble's* threshold to the 49% quantile scored 0.7552 (+0.16), with accuracy, precision, recall and macro-F1 all equal to three decimals, the signature of a matched operating point. Retraining the ensemble with balanced oversampling barely changed its ranking (Spearman 0.946 against the original) and a rank-averaged 18-model variant scored *lower* (0.7381): under this shift, the representation transfers; only the threshold does not. A system tuned to the development prior pays up to 16 macro-F1 points for one constant.

Subtask	Ours	Rank	Top score
A2 fine-grained meme	0.4194	**1st**	0.4194
A1 binary meme	0.7889	3rd	0.823
B2 harm domain	0.7695	3rd	0.7903
B1 safe/unsafe	0.7552	7th	0.9840

Table 2: Official blind-test macro-F1 and final ranks (leaderboards of 19 Aug 2026). Field sizes: 16 (A1), 11 (A2), 22 (B1), 17 (B2) systems. Official baselines: 0.678 (A1), 0.295 (A2), 0.592 (B1), 0.687 (B2).

5 Experimental Results

Table 2 summarizes official results. The A2 trajectory decomposes almost entirely into decode changes: 0.2482 (global threshold) → 0.2690 (per-class + floors) → 0.3650 (VLM, dev-fitted thresholds) → 0.3694 (rank decode) → 0.3913 (5-seed average) → 0.4167 (support-corrected) → 0.4194 (8 seeds). Model-side interventions (seed averaging and 14% more training data) contributed +0.022 and +0.003 respectively; calibration contributed the remaining ∼0.15. The official Task A ranking reinforces the decode thesis: both teams that outrank us on A1 (0.823 and 0.790 vs. our 0.789) trail us on A2 (0.390 and 0.360 vs. 0.419), so the fine-grained ranking tracked decode structure rather than binary-model strength.

On B1, the two systems above 0.98 sit six points clear of the next system (0.924), suggesting an approach qualitatively different from threshold calibration. Engineering notes for reproducibility: a torch 2.9.x Conv3d regression makes the Qwen3-VL vision tower ∼76× slower (13.76s vs. 0.18s per two-image step after pinning torch 2.10); PEFT with gradient checkpointing silently zeroes LoRA gradients unless input gradients are enabled, which we guard with a hard assertion at step 0. Total compute: ≈25 H100-hours, ≈$110.

6 Benchmark Audit and Error Analysis

Sequential-id leakage (Task B, dev release). Sorting the 19,240 released rows by integer id yields seven perfectly contiguous granular-label runs (e.g. adult_content occupies ids 0–6560) and seven binary runs; the two malformed empty-label rows sit exactly on run boundaries, fingerprinting per-domain concatenation before splitting. A nearest-known-id lookup fitted on train only, reading zero characters of text, scores macro-F1 **1.0000** on dev for both B1 and B2; shuffling ids collapses it to 0.51/0.15. Of the 4,813 held-out ids (those in $[0, 24050]$ absent from the released files,

the blind-test pool at release time), 99.7% resolve unambiguously. We identified this on 29 July, before the test release, and did not act on it: we submitted no id-derived predictions. The released test files are re-indexed from 1 and independent of the development ids, so the artifact does not affect evaluation-phase scores. We report it here, with a reproduction script in our code release, so that future releases can block-randomize or hash ids before splitting.

Development sets overstate. The dev-to-test drops (B1 0.954→0.755, B2 0.906→0.770) partly reflect construction: 169 near-duplicate prompt pairs (Jaccard $\geq$0.8, 100% label agreement) span train and dev. Grouped cross-validation shifts measured dev scores by $\leq$0.3 points (the standard alarm overstates), but the test distribution shift does not appear in dev at all.

Feedback as a calibration channel. Per-class scoring output reveals gold support (§4.2). No task rule addresses submission-time use of scoring feedback; we used it openly within the posted submission limits, disclose it here, and report the no-feedback A2 score (0.3913) alongside the final 0.4194 so systems remain comparable. We recommend that future editions aggregate per-class diagnostics or withhold them until the phase closes.

Representative errors. A1 errors concentrate in sarcastic-but-hateful memes where the hateful reading requires cultural context absent from the image; A2 confusions appear dominated by Mocking↔Contempt and Humor↔Sarcasm, mirroring the annotation taxonomy's own soft boundaries. B2's residual errors concentrate in `harassment` vs. `Bully`, categories whose definitions overlap for second-person insults.

7 Conclusion

Conventional models with careful decoding placed 1st of 11 systems on fine-grained meme classification (A2), 3rd of 16 on A1, 3rd of 17 on B2, and 7th of 22 on B1. The general lesson for long-tail safety classification is that macro-F1 is largely a property of the *decision layer*: thresholds must be fitted on the scores actually used at inference, transported to the target prior, and floored on rare classes; and benchmark artifacts (id leakage, latent taxonomy structure, distribution shift) can dominate modeling effort in both directions. Our concurrent ImageEval 2026 systems reach the same conclusion for judge-weight calibration in multimodal evaluation (Younis, 2026). Future work: agreement-aware training against adjudicated test protocols, and organizer-side randomization checks as release practice.

Limitations

Our decode analysis is measured on one task pair and 500-row test sets, where 0.02 macro-F1 $\approx$ 5–10 examples; rank claims between adjacent systems are within seed noise. The B1 balance estimate (49%) is inferred from scoring feedback, not released statistics. Our systems are Arabic-specific in data but not in method; we did not evaluate cross-lingual transfer. The support-corrected A2 decode depends on per-class feedback that other participants may not have used, which complicates strict system-to-system comparison.

Ethical Considerations

Both tasks involve intentionally offensive content; we handled it for research evaluation only, did not redistribute images or prompts, and used the data under its CC BY-NC terms. Better harmful-content classifiers can be repurposed for censorship; per-dialect behavior differences (train safe-rates range from 9.2% for Egyptian to 30.1% for Syrian Arabic) risk unequal moderation quality across dialect communities and should be audited before deployment. We did not exploit the identified label leak, and we document it here with a reproduction so that future releases can protect the integrity of all participants' results.

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